



Palo Alto Networks – URL Filtering Configuration on a PA-220 Palo Alto Next Generation Firewall's Web Graphic User Interface (GUI)

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Purpose

This guide/analysis is meant for personnel who have a PA-220 Next Generation Firewall and need a step by step guide on how to set up a firewall for configuring the access for certain websites in a school, corporate or any other type of network for all users who use the network that the firewall belongs to. This guide also shows basic examples of how to apply the various levels of access on real world used sites.

Background

URL filtering is a security function that can be enabled by firewalls like the PA-220 Next Generation Firewall, which controls what specific websites users on the network can visit depending on the web address they try to access.

For this analysis/guide, we are using the PA-220 Next Generation Firewall and because this firewall is from Palo Alto the URL filtering feature uses a combination of cloud-based threat intelligence and machine learning to analyze and enforce rules on web traffic in real time. The Palo Alto Firewall uses a category-based system where every website in the Palo Alto System is assigned to 1 or more categories like “Malware,” “News,” “Social Networking” etc. This system can be used in PAN-OS, Strat Cloud Manager, or Panorama. These categories organize the websites based on the website content and safety.

The main prioritization is site content which is organized using a unique system to Palo Alto Networks. The Palo Alto Networks database aka PAN-DB is searched whenever a user requests for a specific URL and the firewall performs local lookups and only queries the cloud when necessary to check if the URL has already been categorized. If so, the URL will then check with the category it belongs to, in order to see if and what restrictions apply to it, before implementing those restrictions. If the URL is not in the PAN-DB, the new and unknown website will undergo a real time analysis by machine learning models to categorize the website into the database.

There are 5 actions that can be applied to the categories that contain the websites acknowledged by the database: block, override, continue, alert and allow. In a school environment especially one for younger students, these actions are chosen to prioritize safety, learning and age appropriate content.

Block out of the 5 actions is the most restrictive as it completely prevents users from accessing the websites that are under the category that has been blocked. The process for how the block action actually works is straightforward. First when a blocked site is requested by a user on the network, the firewall will immediately deny the connection and display a status message explaining that access to that site is not permitted along with a custom message that the network administrator can configure on the GUI of the firewall. For a school wide network, the block action will almost always be applied to Malware, Phishing, Gambling, Hacking and Weapons. Besides setting a block action to these categories for the student’s protection it is critical even for a company wide network to set these as blocked sites due to the fact that malware infections which could compromise the network and

device that is used to access these sites have shown to rapidly decrease with this configuration according to Palo Alto.

Override is also an action that will attempt to block the user from accessing the site similar to the Block action but with a key difference. Override can allow access to the website but only after the user or other personnel enters specific credentials that allow access to bypass the block. These credentials in a school wide network would be usually restricted to only teachers or administrators rather than the students themselves who could misuse the access credentials. Some examples of categories that would be set to Override in a school wide network would be Social Networking, Video Streaming, or Blogs and Forums from the PA-220 Next Generation Firewall.

The Continue action is considered out of the 5 possible actions as the least efficient solution to stopping users to access certain websites for the PA-220 Palo Alto Next Generation Firewall. This action only presents users with a warning page but then the user can still access the page if they want to. For school wide networks as well as corporations this level of action is highly inefficient and most likely will never be used. This action if used however, will most likely apply to categories such as Blogs and Forums or General Interest.

The Alert action allows access to a website while generating a log entry each time the site is visited. This action although it does not prevent users from accessing certain sites that network administrators do not want users to access it can help monitor who and for how long a user accessed what sites which can be used for legal purposes if that information is needed later on. This action in a school based network would be most likely applied to categories such as News, General Internet or Search Engines.

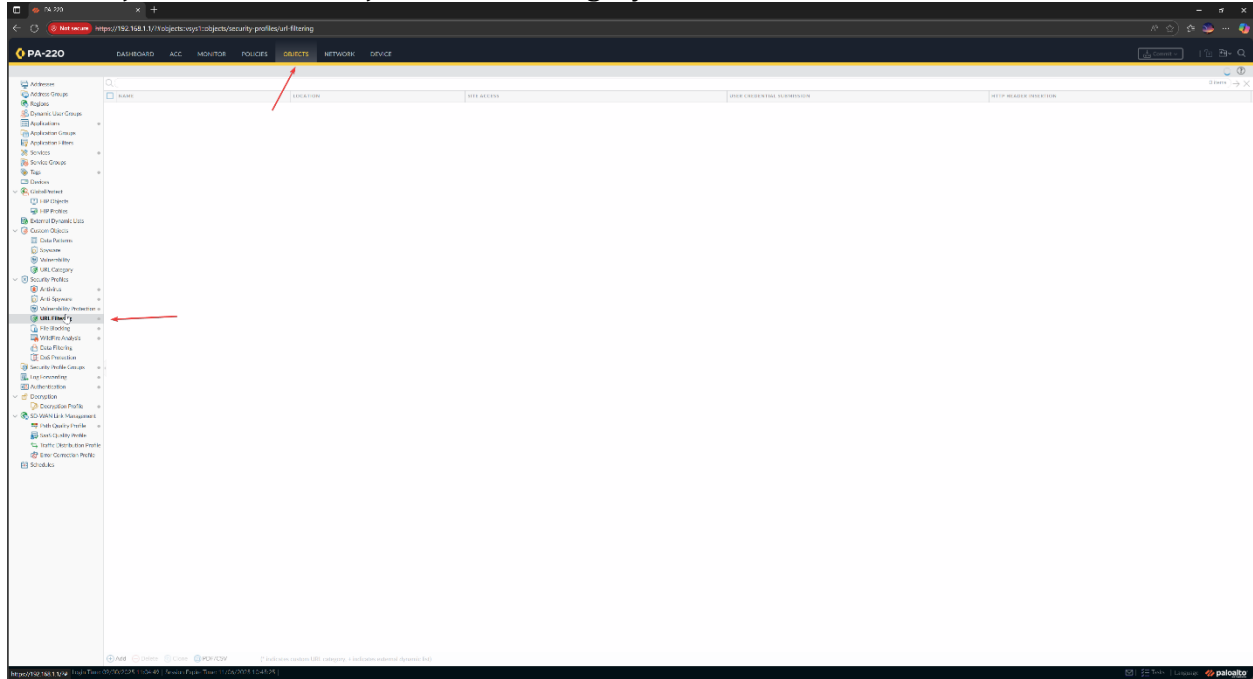
The Allow action is the most likely to be used action out of the 5 in a general school based or corporate network. This action lets any categories that it is assigned to allow the user to continue without any warning or monitoring. For example, School and District Websites, and Online Learning Platforms be freely accessed and without any warning or monitoring. Traffic is normally logged as done with any network regardless of the action assigned to a category and the categories that have been assigned to Allow in terms of a school based network ensures that students can freely use the sites to further their learning and access teacher approved content.

Lab Summary

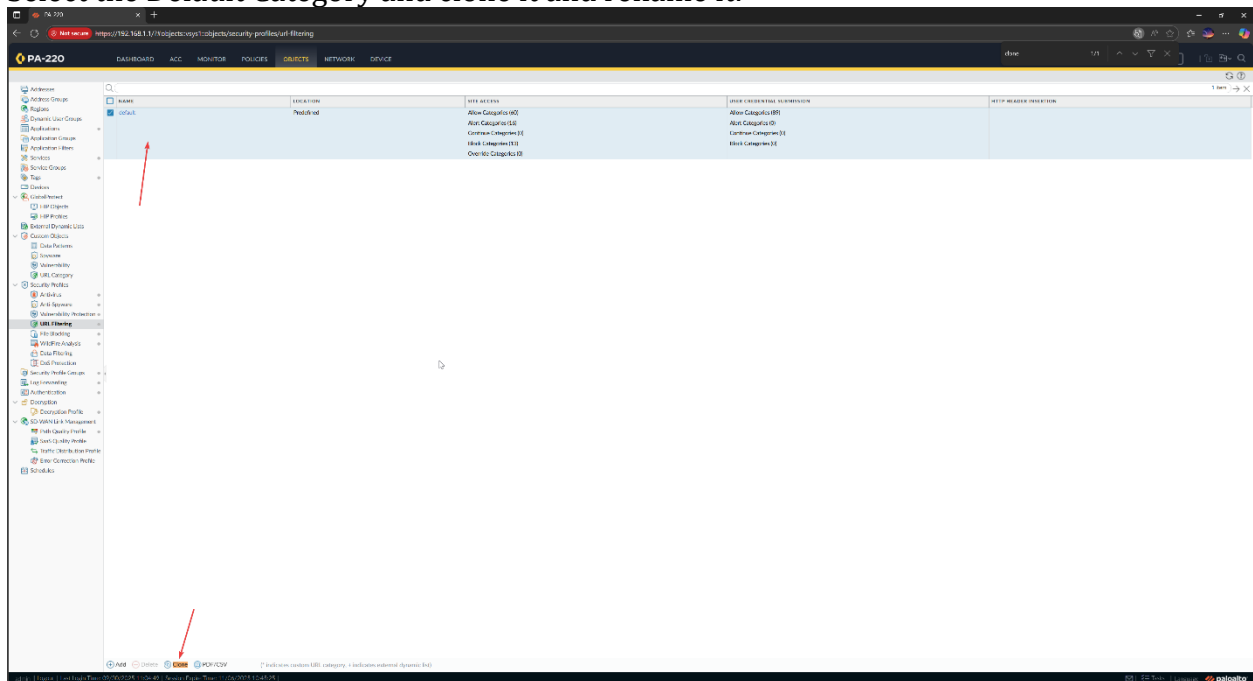
In this analysis, after setting up basic configurations for URL Filtering for websites on a network, we presented a step by step guide on how to properly configure each category containing real world websites to any of the 5 possible actions that can be assigned by the network administrator.

Configurations

Go to Objects > Custom Objects > URL Category.



Select the Default Category and clone it and rename it.



Name your profile and give it a description. Click the check next to your custom URL. For this lab I named my category RealURLFiltering. Click into it

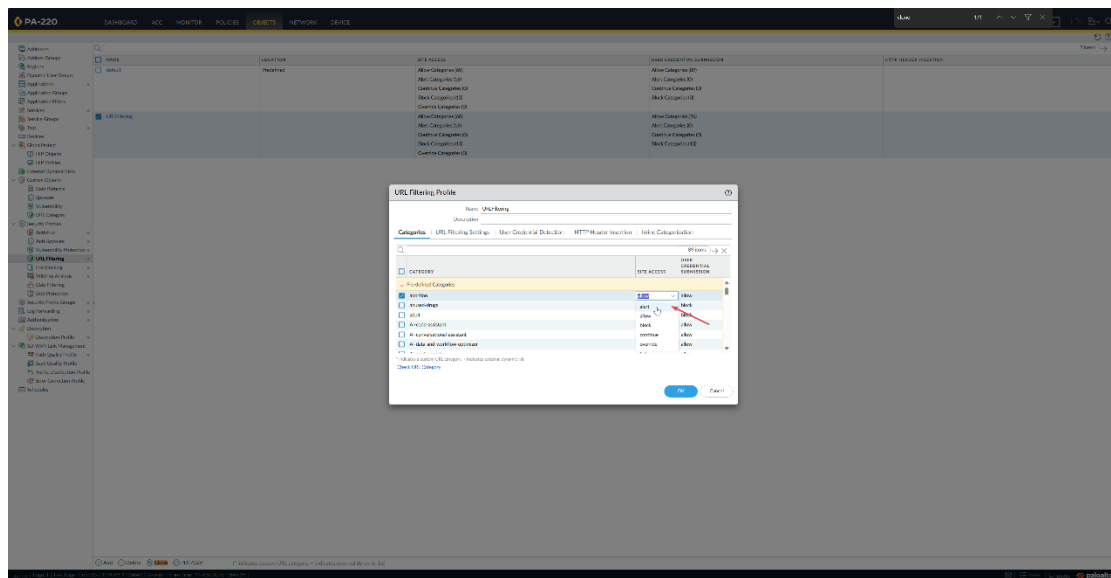
The screenshot shows the Palo Alto Networks PA-220 configuration interface. The left sidebar displays a tree view of configuration objects, with 'URL Filtering' selected under the 'Security Profiles' section. The main pane shows the 'URL Filtering' configuration table. A red arrow points to the 'RealURLFiltering' entry in the table.

NAME	LOCATION	SITE ACCESS	USER CREDENTIAL SUBMISSION	HTTP HEADER PROTECTION
Default	Realtime	Allow Categories (40) Allow Categories (40) Content Categories (3) Block Categories (24) Override Categories (3)	Allow Categories (40) Allow Categories (40) Content Categories (3) Block Categories (24) Override Categories (3)	
URL Filtering		Allow Categories (40) Allow Categories (40) Content Categories (3) Block Categories (24) Override Categories (3)	Allow Categories (40) Allow Categories (40) Content Categories (3) Block Categories (24) Override Categories (3)	
RealURLFiltering		Allow Categories (40) Allow Categories (40) Content Categories (3) Block Categories (24) Override Categories (3)	Allow Categories (40) Allow Categories (40) Content Categories (3) Block Categories (24) Override Categories (3)	

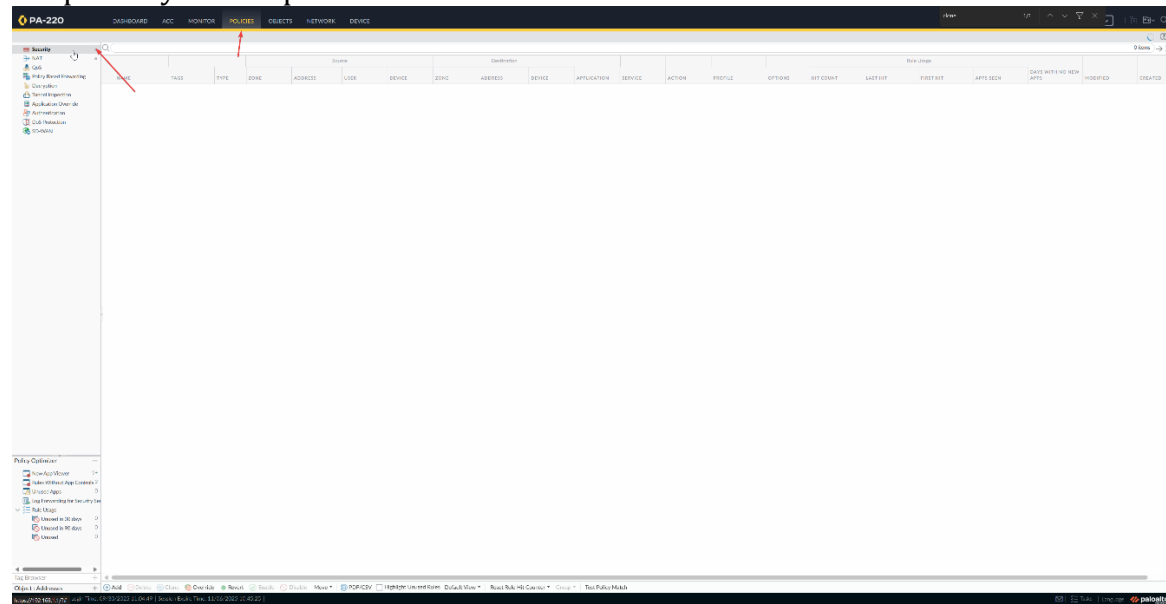
The bottom of the interface shows the 'Add', 'Delete', 'Clone', and 'Refresh' buttons, along with the 'PA-220' model identifier.

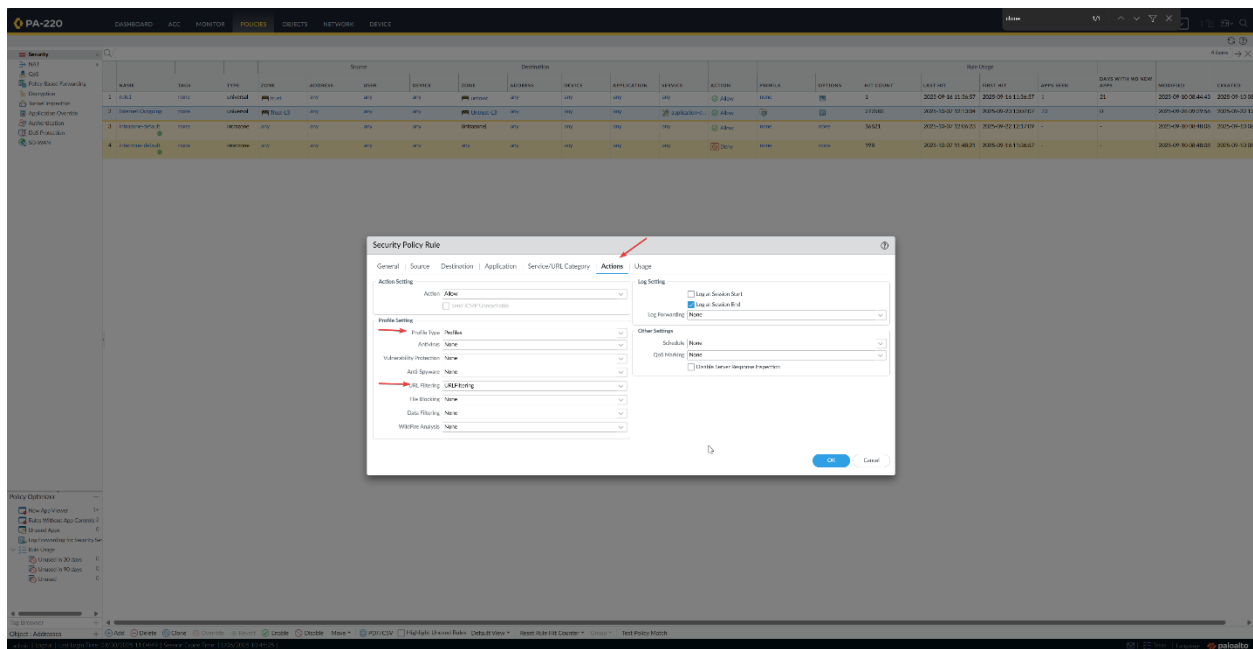
In the section that lists all URL categories, select all categories and then de-select malware, command-and-control, and phishing.

To the right of the Action column heading, mouse over and select the down arrow and then select Set Selected Actions and choose alert. Choose the appropriate action to match expectations for an Elementary School. Make sure "Site Access" and "User Credential Submission" are set to "block". Optionally, you can set the "Site Access" key to "override", which will make the website require an administrator password to access. Properly configure the pre-defined categories to match what would be appropriate for a school setting



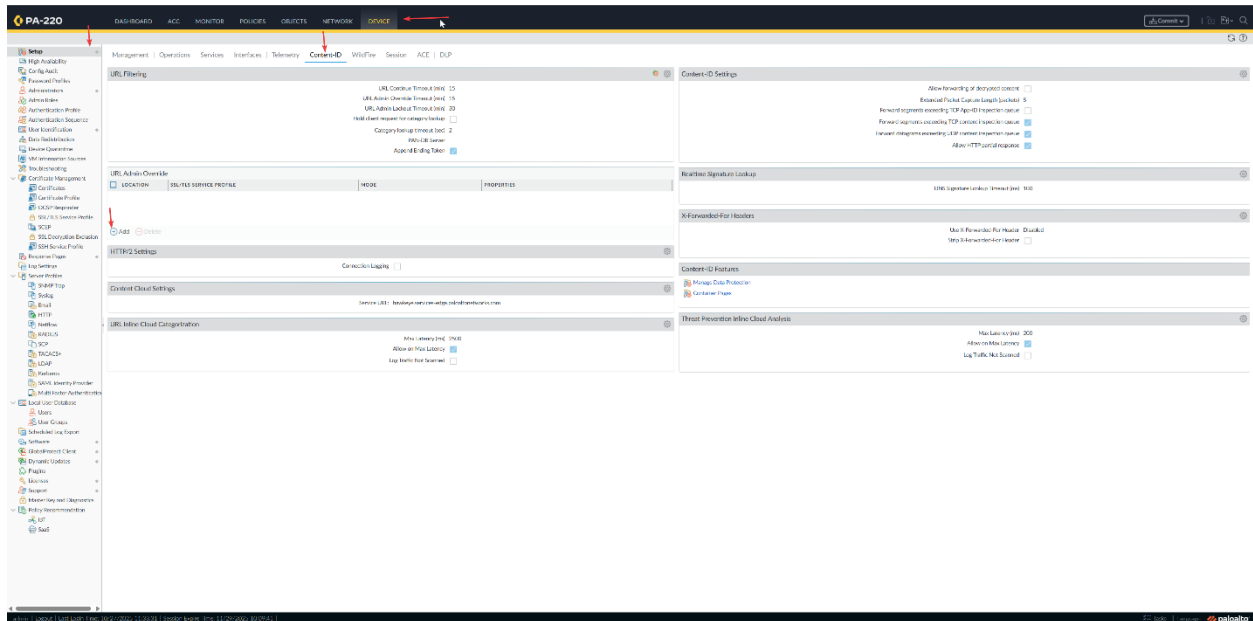
To apply the Url-filtering:
Click the “Actions” tab, set “Profile Type” to “Profiles”, and set the URL Filtering profile to the profile you set up

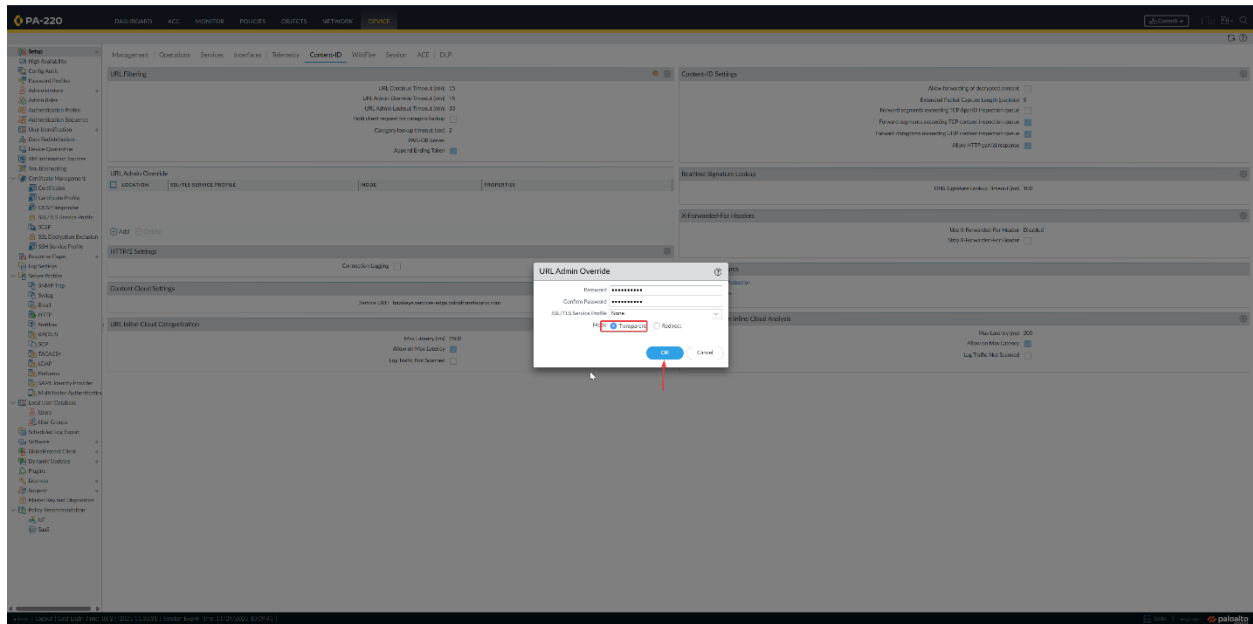




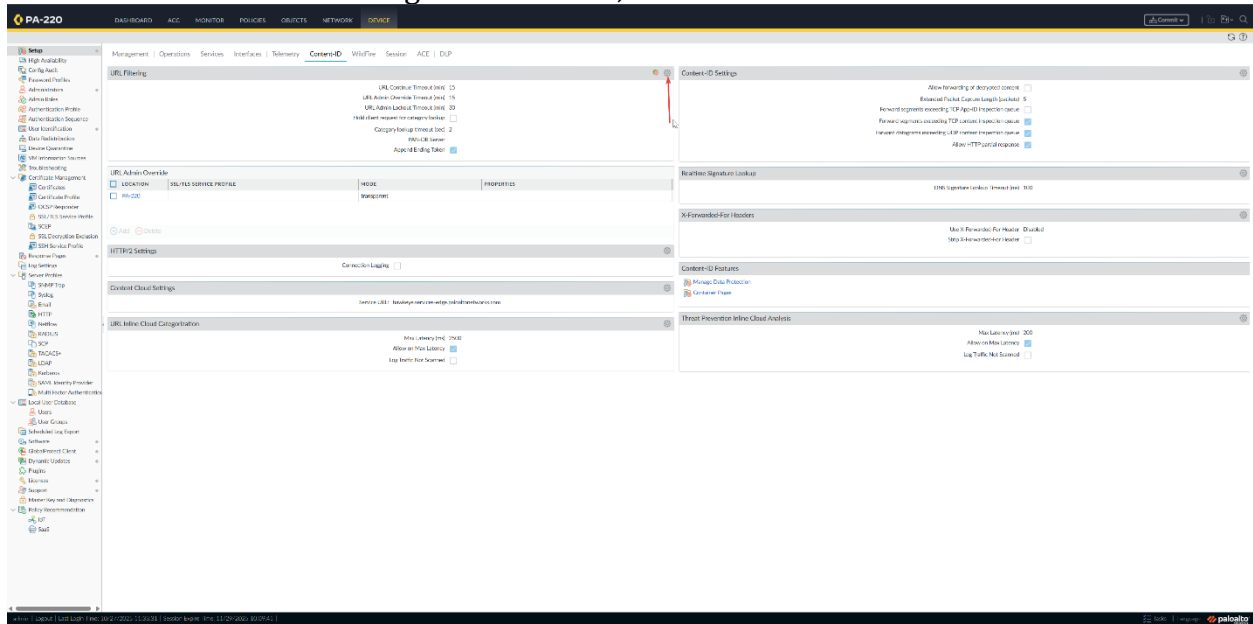
To configure Admin Override:

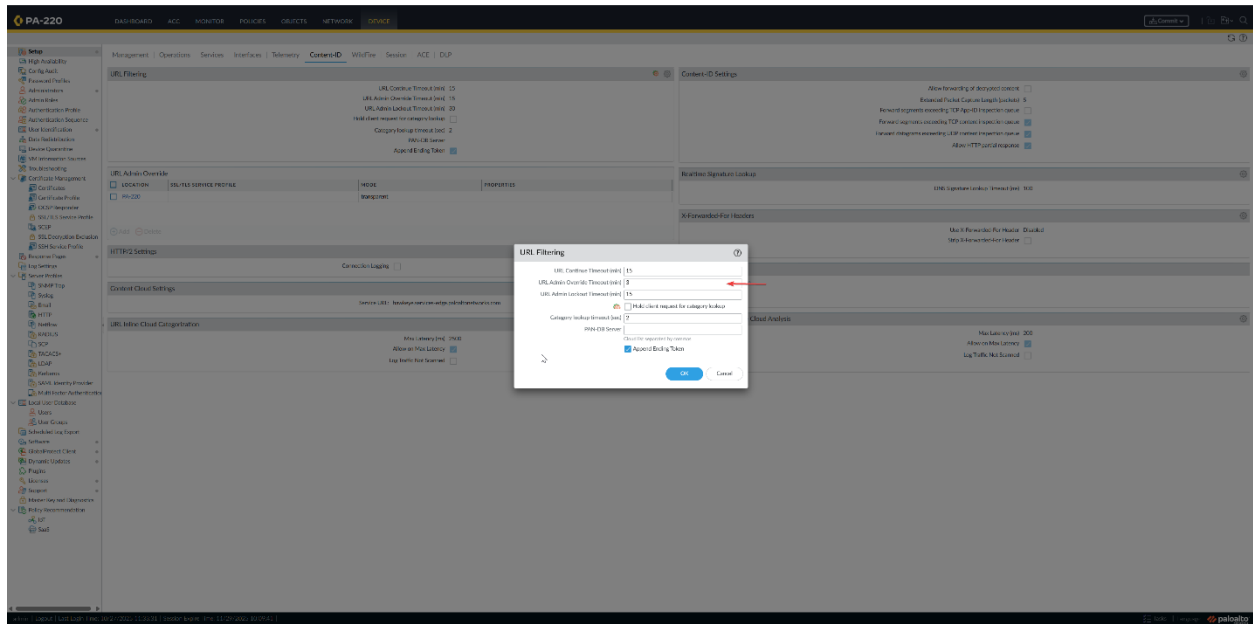
Go to “Device” > “Setup” > “Content ID” and click “Add” under the “URL Admin Override” section. Add your desired password, select transparent as your mode and click “OK”



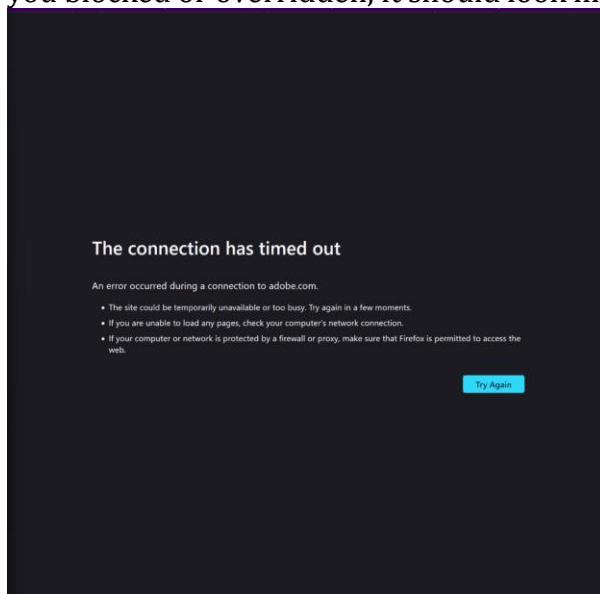


In the device tap click the settings icon on the URL filtering box and change the URL Admin Override Timeout to something lower than 15, I chose 3.

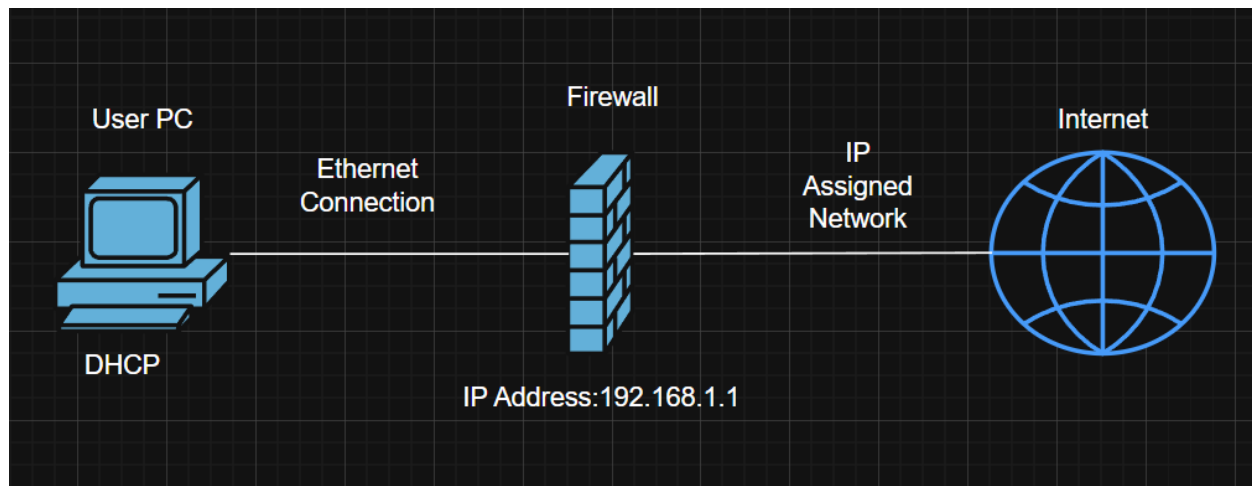




Doing that should work and when you search up any websites in the categories in which you blocked or overridden, it should look like this:



Network Diagram



Problems

One of the first problems I had was when I was configuring the URL filtering security policy to the override action for certain categories. The problem was the fact that when configuring the URL admin override key, I was testing the functionality of our configurations with Amazon however amazon.com was consistently blocked with no override capabilities and after trying to troubleshoot our configurations I realized the fact that Amazon is considered under a different category specifically Shopping than the category that I configured for override. The other problem that I had was during the creation of the SSL/TLS profile section and configure it for Override I initially the configurations were not working but it turns out I just had to use Firefox instead of chrome for the changes to be properly synched.

Conclusion

In this guide/analysis, I presented a step by step guideline on how to set up basic network URL Filtering settings on the Palo Alto Next Generation Firewall specifically PAN-OS 12.1. In this guide I also showed examples of using URL Filtering on real world websites like Amazon.com and explained how each of the possible 5 actions that this firewall can apply, are able to be applied to various websites.